Demo document — not a scientific publication

FrameLite: lightweight clip sampling for on-device action recognition

A. Demo, B. Example — Institute of Fictional Computing

Abstract
We present FrameLite, a clip sampling scheme for action recognition on embedded hardware. FrameLite is trained on the DemoKinetics-200 training split and uses a MobileViT-S backbone with supervised labels only.

1. Method

The backbone is MobileViT-S, pretrained on DemoImage-1k and fine-tuned end to end. We sample 16 frames per clip at an input resolution of 224x224 pixels.

3. Results

On the ED-Bench validation split FrameLite reaches 81.4% top-1 accuracy at 16 frames per clip. We did not evaluate on the held-out test split.

Latency was measured on a single Jetson Orin NX at 34 ms per clip with INT8 quantisation enabled.

4. Limitations

The authors note that accuracy was measured on the validation split only, and that INT8 quantisation costs roughly 0.6 points of accuracy compared with the FP16 model.